1. Technical Assignment (TA)

Project Title: EduOps

Developer: Adilkan Anarbekov

Tech Stack: Flutter (Frontend), Firebase (Backend), C++ (Scheduling Core)

I. Objective

To develop a cross-platform educational management system that automates the generation of optimized, conflict-free academic timetables.

II. Core Features

- **User Roles:** Admin (Principal/Deans), Teachers, and Students.
- **Data Management:** Entry of teachers, subjects, room capacities, and class sizes.
- **C++ Scheduler Engine:** A standalone module using a **Genetic Algorithm** or **Constraint Satisfaction** to generate schedules based on constraints:
  - **Hard Constraints:** No teacher/room double-booking, room capacity limits.
  - **Soft Constraints:** Teacher preferences, avoiding "gaps" in student schedules.
- **Real-time Sync:** Integration with Firebase Firestore for live data updates.

III. Non-Functional Requirements

- **Performance:** Schedule generation should take less than 60 seconds for a mid-sized school.
- **Cross-Platform:** Must run on Android, iOS, and Web (Flutter).
- **Scalability:** Support for multiple departments and complex elective systems.

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2. Project Proposal

To: Alatoo International University / Project Stakeholders

Subject: Proposal for **EduOps** – Next-Generation Automated Academic Scheduling

Introduction

Manual scheduling in educational institutions is a time-consuming, error-prone process. **EduOps** aims to solve this by providing a unified mobile and web platform that automates the creation of complex timetables.

The Solution

Unlike traditional tools, EduOps leverages the speed of **C++** to handle the mathematical complexity of scheduling. This engine is wrapped in a modern **Flutter** interface, providing a user-friendly "EduPage" style experience accessible from any device.

Benefits

- **Time Efficiency:** Reduces scheduling time from days to minutes.
- **Conflict Resolution:** 100% elimination of human errors like double-booked rooms.
- **Resource Optimization:** Maximizes the use of available classrooms and faculty time.
- **Ease of Use:** Cloud-based storage via **Firebase** ensures that teachers and students see updates instantly.

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3. Auto-Schedule Samples & Resources

Here are 5 resources and repositories that provide logic, code, or research samples for C++ scheduling algorithms:

1. **btzy / autotimetable (GitHub):** A high-performance automatic timetable generation engine written in C++.
2. **Solving Timetable Problem with Genetic Algorithms (Research):** Detailed logic on how to implement the C++ core using 3D structures (Days/Slots/Rooms).
3. **Ilia-Abolhasani / scheduler (GitHub):** A genetic algorithm approach specifically for university classroom scheduling.
4. **Genetic Algorithm for School Timetabling (StackOverflow Discussion):** Professional breakdown of ranking constraints in C++.
5. **University Scheduling using GA (Diva Portal):** A comprehensive thesis with C++ logic and benchmark analysis.